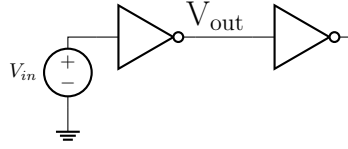
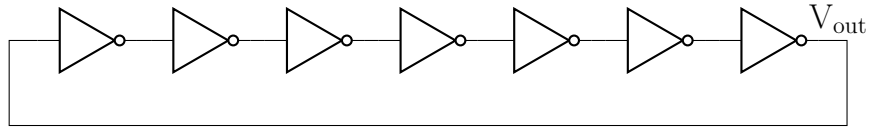


EE5311 Digital IC Design: Assignment 4
Inverter and Ring oscillator – post-layout extracted simulation

1. Draw a DRC and LVS clean layout of the CMOS reference inverter $\left(\frac{W_p}{L_p} = \frac{0.84\mu\text{m}}{0.15\mu\text{m}}, \frac{W_n}{L_n} = \frac{0.42\mu\text{m}}{0.15\mu\text{m}}\right)$.
 - (a) Extract the parasitic values from the layout and find the delay of the inverter using the circuit below. Use the extracted netlist for both the inverter instances.



- (b) Measure the delay including the layout parasitic of net V_{out} .
2. Using the inverter layout above, draw the layout of a seven stage ring oscillator shown below.



To ensure oscillation in the transient simulation, set the node $V_{out} = 0\text{V}$ initially using: `.ic v(Vout)=0`

- (a) Measure the oscillating frequency for $V_{DD} = 1.8\text{V}$ with the layout parasitics.
- (b) Plot the oscillating frequency and time period as a function of V_{DD} for $V_{DD} = 1\text{V}$ to 1.8V in steps of 0.1V .
- (c) Compare the frequencies against pre-layout simulation in Assignment 3.